

```
!pip install streamlit pyngrok nltk
```

```
Requirement already satisfied: streamlit in /usr/local/lib/python3.12/dist-packages (1.57.0)
Requirement already satisfied: pyngrok in /usr/local/lib/python3.12/dist-packages (8.1.2)
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: altair!=5.4.0,!5.4.1,<7,>=4.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (5.5.0)
Requirement already satisfied: blinker<2,>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (1.9.0)
Requirement already satisfied: cachetools<8,>=5.5 in /usr/local/lib/python3.12/dist-packages (from streamlit) (6.2.6)
Requirement already satisfied: click<9,>=7.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (8.3.3)
Requirement already satisfied: gitpython!=3.1.19,<4,>=3.0.7 in /usr/local/lib/python3.12/dist-packages (from streamlit) (3.1.47)
Requirement already satisfied: numpy<3,>=1.23 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.0.2)
Requirement already satisfied: packaging>=20 in /usr/local/lib/python3.12/dist-packages (from streamlit) (26.1)
Requirement already satisfied: pandas<4,>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.2.2)
Requirement already satisfied: pillow<13,>=7.1.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (11.3.0)
Requirement already satisfied: pydeck<1,>=0.8.0b4 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.9.2)
Requirement already satisfied: protobuf<8,>=3.20 in /usr/local/lib/python3.12/dist-packages (from streamlit) (5.29.6)
Requirement already satisfied: pyarrow>=7.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (18.1.0)
Requirement already satisfied: requests<3,>=2.27 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.32.4)
Requirement already satisfied: tenacity<10,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (9.1.4)
Requirement already satisfied: toml<2,>=0.10.1 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.10.2)
Requirement already satisfied: typing-extensions<5,>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (4.15.0)
Requirement already satisfied: starlette>=0.40.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.52.1)
Requirement already satisfied: uvicorn>=0.30.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.46.0)
Requirement already satisfied: httpxtools>=0.6.3 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.7.1)
Requirement already satisfied: anyio>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (4.13.0)
Requirement already satisfied: python-multipart>=0.0.10 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.0.26)
Requirement already satisfied: websockets>=12.0.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (15.0.1)
Requirement already satisfied: itsdangerous>=2.1.2 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.2.0)
Requirement already satisfied: watchdog<7,>=2.1.5 in /usr/local/lib/python3.12/dist-packages (from streamlit) (6.0.0)
Requirement already satisfied: PyYAML>=5.1 in /usr/local/lib/python3.12/dist-packages (from pyngrok) (6.0.3)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from altair!=5.4.0,!5.4.1,<7,>=4.0->streamlit) (3.1.4)
Requirement already satisfied: jsonschema>=3.0 in /usr/local/lib/python3.12/dist-packages (from altair!=5.4.0,!5.4.1,<7,>=4.0->streamlit) (4.23.0)
Requirement already satisfied: narwhals>=1.14.2 in /usr/local/lib/python3.12/dist-packages (from altair!=5.4.0,!5.4.1,<7,>=4.0->streamlit) (1.42.1)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio>=4.0.0->streamlit) (3.13)
Requirement already satisfied: gitdb<5,>=4.0.1 in /usr/local/lib/python3.12/dist-packages (from gitpython!=3.1.19,<4,>=3.0.7->streamlit) (4.0.11)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas<4,>=1.4.0->streamlit) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas<4,>=1.4.0->streamlit) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas<4,>=1.4.0->streamlit) (2025.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.27->streamlit) (3.4.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.27->streamlit) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.27->streamlit) (2025.11.11)
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.12/dist-packages (from uvicorn>=0.30.0->streamlit) (0.16.0)
Requirement already satisfied: smmap<6,>=3.0.1 in /usr/local/lib/python3.12/dist-packages (from gitdb<5,>=4.0.1->gitpython!=3.1.19,<4,>=3.0.7->streamlit) (5.0.6)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2 in /usr/local/lib/python3.12/dist-packages (from altair!=5.4.0,!5.4.1,<7,>=4.0->streamlit)) (3.0.2)
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!5.4.1,<7,>=4.0->streamlit) (25.1.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!5.4.1,<7,>=4.0->streamlit) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!5.4.1,<7,>=4.0->streamlit) (0.36.0)
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!5.4.1,<7,>=4.0->streamlit) (0.25.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas<4,>=1.4.0->streamlit) (1.17.0)
```

```
from google.colab import files
uploaded = files.upload()
```

No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving True.csv to True.csv

```
import pandas as pd
import nltk
import re
import pickle

from nltk.corpus import stopwords
from nltk.stem import PorterStemmer

from sklearn.model_selection import train_test_split

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.naive_bayes import MultinomialNB

from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
```

```

# download stopwords
nltk.download("stopwords")

# NLP objects
stemmer = PorterStemmer()

stop_words = set(
    stopwords.words(
        "english"
    )
)

# load data
fake_df = pd.read_csv(
    "Fake.csv"
)

true_df = pd.read_csv(
    "True.csv"
)

# labels
fake_df["label"] = 0

true_df["label"] = 1

# combine
df = pd.concat(
    [
        fake_df,
        true_df
    ]
)

# keep text only
df = df[
    [
        "text",
        "label"
    ]
]

df = df.dropna()

# shuffle
df = df.sample(
    frac=1,
    random_state=42
)

# NLP preprocessing
def preprocess_text(text):

    text = text.lower()

    text = re.sub(
        r"^[^a-zA-Z]",
        "",
        text
    )

    words = text.split()

    processed_words = []

    for word in words:

        if word not in stop_words:

            stem_word = stemmer.stem(
                word
            )

            processed_words.append(
                stem_word
            )

```

```
        return " ".join(
            processed_words
        )

# clean
df["text"] = df[
    "text"
].apply(
    preprocess_text
)

X = df["text"]

y = df["label"]

# split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# TF-IDF
vectorizer = TfidfVectorizer(
    max_features=20000,
    ngram_range=(1,2)
)

X_train_vec = vectorizer.fit_transform(
    X_train
)

X_test_vec = vectorizer.transform(
    X_test
)

# model
model = MultinomialNB()

# train
model.fit(
    X_train_vec,
    y_train
)

# predict
y_pred = model.predict(
    X_test_vec
)

print(
    "Accuracy:",
    accuracy_score(
        y_test,
        y_pred
    )
)

print(
    classification_report(
        y_test,
        y_pred
    )
)

# save
pickle.dump(
    model,
    open(
        "model.pkl",
        "wb"
    )
)
```

```

pickle.dump(
    vectorizer,
    open(
        "vectorizer.pkl",
        "wb"
    )
)

print(
    "MODEL SAVED"
)

```

[nltk_data] Downloading package stopwords to /root/nltk_data...

[nltk_data] Package stopwords is already up-to-date!

/tmp/ipykernel_499/1509599836.py:31: DtypeWarning: Columns (4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28

fake_df = pd.read_csv(

Accuracy: 0.9518032056990204

	precision	recall	f1-score	support
0	0.95	0.96	0.95	4701
1	0.95	0.95	0.95	4283
accuracy			0.95	8984
macro avg	0.95	0.95	0.95	8984
weighted avg	0.95	0.95	0.95	8984

MODEL SAVED

Double-click (or enter) to edit

```

%%writefile app.py

import streamlit as st

import pickle

import nltk

import re

from nltk.corpus import stopwords

from nltk.stem import PorterStemmer

nltk.download(
    "stopwords"
)

stemmer = PorterStemmer()

stop_words = set(
    stopwords.words(
        "english"
    )
)

# load files
model = pickle.load(
    open(
        "model.pkl",
        "rb"
    )
)

vectorizer = pickle.load(
    open(
        "vectorizer.pkl",
        "rb"
    )
)

# NLP preprocessing
def preprocess_text(text):

    text = text.lower()

```

```

text = re.sub(
    r"[^a-zA-Z]",
    " ",
    text
)

words = text.split()

processed = []

for word in words:

    if word not in stop_words:

        processed.append(
            stemmer.stem(
                word
            )
        )

return " ".join(
    processed
)

st.set_page_config(
    page_title="Fake News Detector"
)

st.title(
    "📰 Fake News Detection using NLP"
)

news = st.text_area(
    "Enter news article"
)

if st.button(
    "Predict"
):

    clean_news = preprocess_text(
        news
    )

    vec = vectorizer.transform(
        [
            clean_news
        ]
    )

    pred = model.predict(
        vec
    )[0]

    probs = model.predict_proba(
        vec
    )[0]

    st.write(
        f"Fake Probability: {probs[0]*100:.2f}%"
    )

    st.write(
        f"Real Probability: {probs[1]*100:.2f}%"
    )

    if pred == 0:

        st.error(
            "❌ FAKE NEWS"
        )

    else:

        st.success(
            "✅ REAL NEWS"
        )

```

Writing app.py

```
from pyngrok import ngrok

ngrok.set_auth_token("3DnIlzcAnBW0aKHjaV6xuMY7vij_7vnd6onKU7niRm8xxktbR")
```

```
!pkill -f streamlit
```

```
!streamlit run app.py --server.port 8501 &>/dev/null &
```

```
from pyngrok import ngrok

ngrok.kill()

public_url = ngrok.connect(
    8501
)

print(
    public_url
)
```

NgrokTunnel: "<https://epilogue-twister-trapped.ngrok-free.dev>" -> "<http://localhost:8501>"

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